

GAP_ANALYSIS_SUMMARY

HelixCode Gap Analysis - Executive Summary

Document: Full analysis at /Users/milosvasic/Projects/helix_code/GAP_ANALYSIS.md **Date:** 2025-11-04

Critical Findings (TL;DR)

What's MISSING (Critical):

1. **Anthropic/Claude Provider** - No direct Claude API integration
2. **Google Gemini Provider** - Missing 1M+ context models
3. **Extended Thinking** - Can't use Claude's reasoning mode
4. **Prompt Caching** - Missing 90% cost savings feature
5. **AWS Bedrock** - No enterprise AWS support
6. **Azure OpenAI** - No Microsoft enterprise support

What's GOOD (Strengths):

1. **7 Providers Already:** OpenAI, Ollama, Llama.cpp, Qwen, xAI, OpenRouter, Copilot
 2. **MCP Protocol:** Full implementation with WebSocket
 3. **Distributed Workers:** Unique SSH-based worker pool
 4. **Task Checkpointing:** Advanced work preservation
 5. **Multi-Platform:** CLI, TUI, Desktop, Mobile
 6. **Reasoning Engine:** Built-in reasoning capabilities
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Priority Roadmap

CRITICAL (Week 1-2): MUST DO FIRST

1. Add Anthropic Provider (claude-3.7-sonnet, claude-4-sonnet)
Location: /internal/llm/anthropic_provider.go
2. Add Gemini Provider (gemini-2.5, gemini-2.5-flash)
Location: /internal/llm/gemini_provider.go

3. Implement Extended Thinking Support
 - Add: ReasoningEffort field to LLMRequest
4. Implement Prompt Caching
 - Add: cache_control markers for cost savings

HIGH (Week 3-4): Do Next

5. AWS Bedrock Provider
 - Location: /internal/llm/bedrock_provider.go
6. Azure OpenAI Provider
 - Location: /internal/llm/azure_provider.go
7. Vision Auto-Switching
 - Like Qwen Code - auto-detect images
8. Context Compression
 - /compress command for long sessions

MEDIUM (Week 5-6): Important

9. VertexAI Provider (Google Cloud)
10. Groq Provider (fast inference)
11. File System Tools (read/write/search)
12. Shell Execution Tools (safe command running)
13. Web Tools (search, fetch)

LOW (Week 7+): Nice to Have

14. VS Code Extension
15. YOLO Mode (auto-approve)
16. Memory System (long-term storage)
17. Enhanced TUI (better interactivity)

Key Metrics

Current State:

- **Providers:** 7 implemented
- **Models:** ~25 models supported
- **Context Sizes:** Up to 1M (Qwen Turbo)

- **Features:** Basic streaming, tool calling, reasoning

Target State (After Phase 2):

- **Providers:** 11+ implemented
- **Models:** 50+ models supported
- **Context Sizes:** Up to 2M (Gemini)
- **Features:** Extended thinking, caching, vision auto-switch

Success Metrics (Week 10):

- Feature parity with Claude Code
- Feature parity with Qwen Code
- Superior provider support vs Goose
- Enterprise-ready (Bedrock + Azure)

Implementation Estimates

Feature	Priority	Effort	Impact
Anthropic Provider	CRITICAL	3-4 days	HUGE
Gemini Provider	CRITICAL	3-4 days	HUGE
Extended Thinking	HIGH	2 days	HIGH
Prompt Caching	HIGH	2 days	HIGH
Bedrock Provider	HIGH	3 days	MEDIUM
Azure Provider	HIGH	3 days	MEDIUM
Vision Auto-Switch	MEDIUM	2 days	MEDIUM
Context Compression	MEDIUM	3 days	MEDIUM

Total Critical Path: ~2 weeks for minimum viable product **Total Feature Complete:** ~10 weeks for full roadmap

Code Locations Cheat Sheet

Where to Add Providers:

```
# New provider files
/Users/milosvasic/Projects/helix_code/helix_code/internal/llm/anthropic_provider.go
```

```
/Users/milosvasic/Projects/helix_code/helix_code/internal/llm/  
    gemini_provider.go  
/Users/milosvasic/Projects/helix_code/helix_code/internal/llm/  
    bedrock_provider.go  
/Users/milosvasic/Projects/helix_code/helix_code/internal/llm/  
    azure_provider.go
```

```
# Update provider enum
```

```
/Users/milosvasic/Projects/helix_code/helix_code/internal/llm/  
    provider.go:17-27
```

```
# Update factory
```

```
/Users/milosvasic/Projects/helix_code/helix_code/internal/llm/  
    provider.go:339-356
```

Existing Provider Examples:

```
# Best reference for new cloud providers
```

```
/Users/milosvasic/Projects/helix_code/helix_code/internal/llm/  
    openai_provider.go  
/Users/milosvasic/Projects/helix_code/helix_code/internal/llm/  
    qwen_provider.go
```

```
# OAuth2 example
```

```
/Users/milosvasic/Projects/helix_code/helix_code/internal/llm/  
    qwen_provider.go:46-98
```

```
# Token exchange example
```

```
/Users/milosvasic/Projects/helix_code/helix_code/internal/llm/  
    copilot_provider.go:65-162
```

Competitive Analysis Summary

Claude Code

- **Strengths:** Native Claude integration, extended thinking
- **HelixCode Gap:** Missing Anthropic provider
- **Action:** Implement Anthropic provider ASAP

Qwen Code

- **Strengths:** Vision auto-switch, context compression, OAuth2
- **HelixCode Gap:** Missing vision auto-switch, compression
- **Action:** Add vision detection + compression commands

Codename Goose

- **Strengths:** Desktop UI, extension system
- **HelixCode Gap:** VS Code extension
- **Action:** Low priority - desktop app already exists

OpenCode

- **Strengths:** 9 providers, LSP support
 - **HelixCode Gap:** Missing Gemini, Anthropic, Bedrock, Azure
 - **Action:** Add missing providers (Phases 1-2)
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Risk Mitigation

High Risk Items:

1. **API Changes:** Anthropic/Gemini APIs evolve
 - **Mitigation:** Follow official SDKs, version lock
2. **Cost Control:** More providers = complex billing
 - **Mitigation:** Token tracking, budget limits, caching
3. **Rate Limits:** Each provider different
 - **Mitigation:** Per-provider rate limiters

Medium Risk Items:

1. **Auth Complexity:** OAuth, AWS, Azure, GCP
 - **Mitigation:** Credential manager abstraction
 2. **Context Windows:** Different limits per model
 - **Mitigation:** Dynamic management
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Why This Matters

Without These Features:

- Can't compete with Claude Code
- Can't serve enterprise AWS customers
- Can't serve enterprise Azure customers
- Missing 90% cost savings from caching
- Can't use best-in-class Claude reasoning

With These Features:

- Most comprehensive provider support
 - Enterprise-ready (AWS, Azure, GCP)
 - Cost-optimized (caching, compression)
 - Best-in-class reasoning (Claude + native)
 - Unique distributed architecture
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Next Steps

1. **Read Full Analysis:** /Users/milosvasic/Projects/helix_code/GAP_ANALYSIS.md
 2. **Start with Anthropic:** Week 1 implementation
 3. **Add Gemini:** Week 2 implementation
 4. **Test & Deploy:** Week 3-4
 5. **Continue Roadmap:** Phases 3-5
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Questions to Ask

1. **Should we prioritize cost savings?**
 - YES → Implement prompt caching in Week 1
 - NO → Focus on provider breadth first
 2. **Do we need enterprise support?**
 - YES → Bedrock + Azure in Phase 2
 - NO → Skip to tools in Phase 3
 3. **Is vision critical?**
 - YES → Vision auto-switch in Week 3
 - NO → Defer to Phase 4
 4. **Do we want VS Code?**
 - YES → Allocate Phase 4 resources
 - NO → Focus on CLI/TUI excellence
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Bottom Line: HelixCode is **2 weeks away** from being competitive with Claude Code and Qwen Code. The architecture is solid, just need to add the cloud providers and advanced features.

Recommended: Start with Anthropic and Gemini providers immediately. Everything else can follow.